MATH5320: Financial Risk & Management (HMWK3)

1. **VaR**

   (a) What is VaR?

   (b) What desirable property or properties of risk measures does it fail to exhibit?

   (c) Give (an) example of the above property failure(s).

1)

a)

The p-level Value at Risk is the smallest number X, such that the losses do not exceed X with probability p. For instance, under

$$VaR_p = inf\ \{X: P(V_0 - V_T \leq X) \geq\ p\}$$

Alternatively, it can be shown as

$$VaR(V, T, P) = G^{-}1(p), such\ that$$
$$G(X) = P(V_0 - V_T \leq X) = E^P\left[1_{V_0 - V_T \leq X}\right],$$

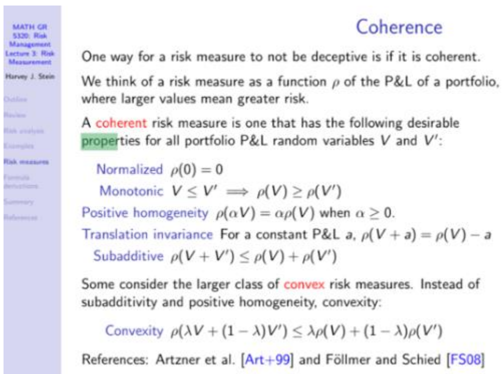
Where it is the p-quantile of the loss-distribution, where final value, or PnL of the position is shown as $V_0 - V_T$. Interpretively, we have a 99% VaR, such that if it is \$1mm, 99% of the time losses are \$1mm or less over the time horizon. In many ways, it is a great risk measure, as it does offer a precise percentile loss, and in turn, it is operationally actionable. As it is "elicitable", it can be back-tested.

b)

In terms of the desirable aspects of risk measures, listed in slide 28, it is **hard to back-test** accurately, and **substantively harder to compute** than the standard deviation, or volatiliy.

Invariably, the condition that can ***fail*** **is the subadditivity** condition for VaR in terms of *coherence*. From the slide 29 given, we have the following conditions of coherence for P&L random variables V & V':

Normalisation condition is satisfied where $\rho(0) = 0$ is satisfied, as if there is no risk, the p-quantile loss is 0. Following, monocity is fulfilled, as larger PnL, means lower risk, and vice versa, where "worse portfolio" means smaller PnL. VaR satisfies monotonicity, because quantiles preserve ordering. Homogeneity is satisfied, as quantiles scale linearly, such that $\rho(\alpha\ V) = \alpha\rho(V)$, for $\alpha \geq\ 0$. Translation invariance is similar, as VaR is shift invariant given that a "sure gain" α, will shift the distribution down by α.



Subadditivity, does fail, as VAR, only looks at one percentile, but not the severity of tail outcomes, and therefore, and can miss how risks combine in the tail, and diversification can increase measured risk (diversification is supposed to reduce risk), and therefore fail the following:

$$\rho(V + V') \leq \rho(V) + \rho(V').$$

Intuitively, two positions, can have losses that are rare, and just below the p-cutoff, such that they have a "rare but nonzero" losses small enough that they occur less than 1- p of the time individually, such that VaR = 0, but when combine, the chance that at least one loss occurs can exceed 1- p, causing the VaR to increase, such that in this case, diversification can actually increase measured risk.

Notably, VaR performs well for certain cases, and is *coherent* and does not fail for elliptically distributed factors, like for multivariate normals, where quantiles behave *nicely* and subadditivity holds. Outside such settings with fat tails, skewness, non-linear payoffs, VaR's lack of general subadditivity is a key problem.

c)

VaR can violate the subadditivity or diversification property

$$\rho(V + V') \leq \rho(V) + \rho(V').$$

This is because, loss positions, can individually be zero, but combined VaR be positive, and therefore undermine the key tenet of portfolio allocation.

In considering the set-up, we generally consider the general situation, where we have $p = 0.95$, where usually it is 0.95/0.99 in industry, and the losses are defined as follows.

$$L := V_0 - V_T$$

Then,

$$L_1 = \begin{cases} 1 \; with \, prob \; 0.04 \\ 0 \; with \, prob \; 0.96 \end{cases}, L_2 = \begin{cases} 1 \; with \, prob \; 0.04 \\ 0 \; with \, prob \; 0.96 \end{cases}$$

Where we assume L_1 & L_2 are independent. Then consider the VaR, of each individually, such that we have it at level p, or the p-quantile of the loss distribution:

$$P(L_1 \leq 0) = P(L_1 = 0) = 0.96 > 0.95,$$

So therefore, in this case the smallest x achieving probability at least 0.95, is x = 0. Hence, the VaR of L_1 & L_2 are 0, individually. Now, we define a combined portfolio, such that:

$$L := L_1 + L_2$$

And here, we consider the probabilities, such that we have "0" loss, as shown:

$$P(L = 0) = P(L_1 = 0, L\_2 = 0) = 0.96 * 0.96 = 0.9216$$

Therefore, we can see that the:

$$P(L \leq 0) = 0.9216 < 0.95$$

As such, the 95% VaR of the total portfolio cannot be 0. Then, checking when x = 1,

$$P(L \leq 1) = 1 - P(L = 2) = 1 - (0.04 * 0.04) = 1 - 0.0016 = 0.9984 \geq 0.95$$

As such, the smallest *x* with $P(L \leq X) \geq 0.95$ is x = 1. As such,

$$\mathrm{VaR}_{0.95}(L_1 + L_2) = 1$$

So, therefore,

$$\mathrm{VaR}_{0.95}(L_1) = 0, \mathrm{VaR}_{0.95}(L_2) = 0, \mathrm{VaR}_{0.95}(L_1 + L_2) = 1$$

Going back, we have:

$$\mathrm{VaR}_{0.95}(L_1 + L_2) = 1 > 0 + 0 = \mathrm{VaR}_{0.95}(L_1) + \mathrm{VaR}_{0.95}(L_2)$$

, so therefore, this violates the definition given as:

$$\rho(V + V') \leq \rho(V) + \rho(V'),$$

when p(.) is VaR applied to losses. To wrap up, each position has a rare chance of a certain loss occurring, and less than the 5% tail probability (1 - p), and once combined, the probability at least some loss occurs is now higher than the 5% tail, such that the quantile loss is higher, and therefore diversification can increase VaR.

2)

(a) What is ES?

(b) What desirable property or properties of risk measures does it fail to exhibit?

a)

P-th percentile Expected shortfall,

$$ES(V, T, p) = -E^P[V_T - V_0 | V_T - V_0 < -VaR(V_T, p)]$$
$$= E^P[V_0 - V_T | V_0 - V_T > VaR(V_T, p)],$$

As defined in slide 40.

The uses the VaR_p threshold, such that with probability p, the loss at most is VaR_p. Then, ES effectively refers to that given the worse 1 – p fraction of outcomes, what is the average loss. In more technical terms, ES measures how large losses are on expectation, conditional on the worst possible 1– p outcomes.

b)

Relative to the coherence properties listed on slide 29, ES does satisfy all the coherence conditions.

- Normalised, $\rho(0) = 0$
- Monotonic, such that worse PnL outcomes imply higher ES

- Positive homogeneity, such that scaling the portfolio, means ES is also scaled proportionally
- Translation means adding a “sure gain’ reduces ES the same rate.
- Sub additive, such that diversification never increases risk.

As such, expected shortfall is a coherent risk measure, and unlike VaR, and is therefore not “deceptive’ in the coherence sense.

It does, however, have some issues. For instance, notably, if the PnL portfolio is normally distributed, ES, functionally becomes something akin to:

$$ES_p = mean + (\text{constant dependent on } p) * STDEV$$

Therefore, **under normality, ranking portfolios becomes equivalent to ranking by volatility**, and ranking portfolios are effectively akin to ranking by volatility, and therefore “the same as variance if distribution is normal”.

It is notably more **computationally expensive** to computed, as it requires estimating and average the tail beyond the quantile, especially in simulation-based environments with limited observations.

Most importantly, however, is the **lack of elicitability** in the validation-front, but it is still back-testable. In human terms, elicitability, refers to the notion of whether a clean way is there to check if the risk forecast is correct using data. A risk measure is ‘elicitable’ if there is a scoring or loss function, $L(x, V)$, such that the true value of $\rho(V)$ is the unique minimiser of the expected score:

$$\rho(V) = arg\ min_x E[L(x, V)]$$

Note: “Scoring or loss function”, basically refers to the idea of when we predict an X (our risk forecast), and when we observe realised V, what is the L(x,V), or error it was, or a “marking scheme” or “error function” like in machine learning.

And the idea is that, if we forecast *x* every day, and then the realised V occurs, and the $L(x, V)$ reports how wrong the forecast was, and elicitability means there exists a “marking scheme” under which the best strategy is to report the true “risk measure”, and that is always unique. In this case, it is not, and therefore ES is not elicitable on its own, as there is no single scoring rule that can “grade” or evaluate ES forecasts. Practically, this makes ES harder to validate with simple statistical tests.

However, **(VaR, ES) together are jointly elicitable**, meaning that there is a loss function that can more accurately validate ES using procedures that involve VaR, and is hence “backtestable”, but ES by itself is not elicitable.

As far as actionability, ES reports an average of extreme losses, than can be more informative than VaR for capital adequacy, it is **less directly actionable**, for example in terms of risk limit.

3)

3. Scenario and sensitivity analysis

Intel is currently trading at \$24.65, interest rates are currently 4.7%, and 1.5 year call options on Intel with a strike of \$25 are trading at an implied vol of 40%.

Suppose you have 1200 shares of Intel.

(a) What is the current value of the position and the sensitivity of your position to moves of Intel?

(b) How many 1.5 year call options struck at \$25 should you write so that your sensitivity to moves of Intel becomes zero?

(c) After writing the calls necessary to eliminate your delta, what are your positions, their market values and the total value of your portfolio?

(d) After putting on the position from the previous part, what would the loss be if the stock moves up 15.0% and what would be the loss if the stock moves down 15.0%

(e) How does the risk of the portfolio with the call options compare to the risk of the portfolio without the call options?

a)

We are only given the 1,200 shares on intel in the current market, so therefore at time 0, we have:

$$V_0 = 1200 \times 24.65 = 29{,}580.$$

If we consider the delta, which is 1 for a stock, we have:

$$\frac{\partial}{\partial S}(1200S) = 1200$$

So a \$1 change in Intel changes the position value by \$1200.

b)

To ensure that sensitivity becomes 0, we need to delta-hedge it. So as such, we need to write n-calls, such that we want to find the *n*, such that we are delta-neutral.

$$\Delta_{\text{total}} = 1200 - n\Delta_{\text{call}}.$$

And the standard result from BSM, is that the delta is:

$$\Delta_c all = \frac{\partial C}{\partial S} = N(d_1), where d_1 = \frac{ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

Plugging in the numbers, from (https://www.barchart.com/options/options-calculator)

Input Parameters

Underlying Price	24.65
Strike Price	25
DTE (days)	547.5
Risk-free rate%	4.7%
Volatility	40%
Dividend Yield% (equities)	0

Calculated Theoretical Values

Theoretical Price	5.35
Delta	0.64061
Gamma	0.03096
Vega	0.11288
Theta	-0.00547
Rho	0.15669

Notably, a manual calculation yields a similar result that is correct to 4d.p with the calculator.

$$\Delta_{\text{total}} = 1200 - n\Delta_{\text{call}}.$$
$$n = \frac{1200}{0.6406} = 1873.2438 \text{ (4. dp)}$$

So, to fully delta hedge, we need about 1873 calls.

c)

We will utilise the options calculator from MATH5010 by Dr Smirnov:
https://www.math.columbia.edu/~smirnov/options13.html

Options Calculator

Calculates Prices of Options

On Divident Paying Stocks

STOCK PRICE:	24.65	NO OF TREE NODES :	20
STRIKE PRICE:	25		
INTEREST RATE 0.1 for 10% :	0.047		
CONT DIV YIELD 0.015 for 1.5%:	0		
VOLATILITY PER YEAR 0.3 for 30% :	0.4		
TIME TO EXPIRATION IN DAYS :	547.5		
AMERICAN PUT PRICE (bin. tree):	4.27348719	Black-Scholes	
EUROPEAN PUT PRICE (bin. tree):	4.07127613	EUR PUT PRICE :	3.99327240
AMERICAN CALL PRICE (bin. tree):	5.42308264	Black-Scholes	
EUROPEAN CALL PRICE (bin. tree):	5.42308264	EUR CALL PRICE :	5.34507891

Calculate | Clear

It shows that the European call price, should be about $5.3451.

So we note that, in practice with n = 1873, and from the BSM calculator, we get \$5.3451, we have the following positions of:

- Long 1200 shares of Intel.
- Short 1873 call options (strike 25, maturity 1.5y).
- Cash received from writing calls.

We consider the market values of the stock, short calls, and cash. In reality, we only consider the stock, as we "sold" the calls. So, the market values are:

$$V_{stock,0} = 1200 \times 24.65 = 29{,}580.$$

$$V_{calls,0} = -1873 \times 5.34507891 = -10{,}011.33$$

$$Cash_0 = +1873 \times 5.34507891 = +10{,}011.33$$

As such, $V_{\text{total},0} = V_{calls,0} + V_{stock,0} + Cash_0 = V_{stock,0} = \$29{,}580,$ as the total portfolio value. Notably, we also have about 0 delta.

d)

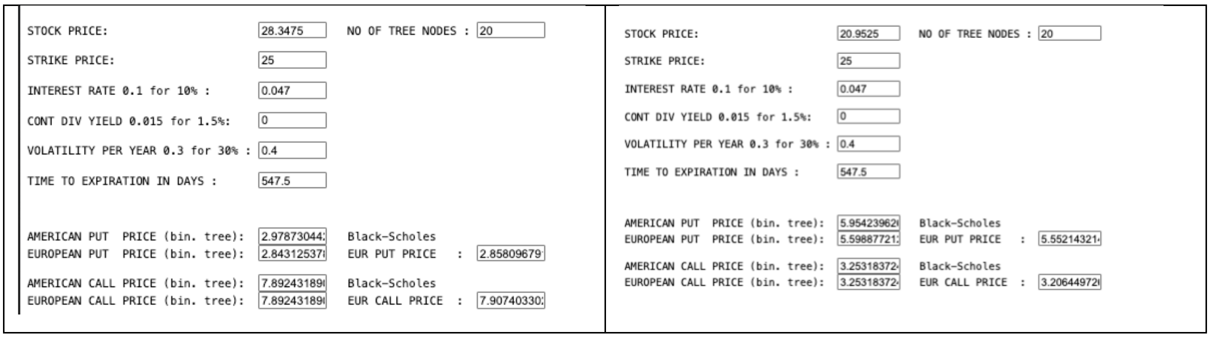
We consider the following scenario analysis of the underlying stock moving up and down 15%. The risk factor of intel stock price has altered. We need to reprice the portfolio under each scenario using the same pricing model of BSM, and identical assumptions into said model.

First considering the stock part of the portfolio:

$$S_{\text{up}} = 24.65(1.15) = 28.3475, \qquad S_{\text{down}} = 24.65(0.85) = 20.9525.$$

Now, the options need to be repriced using the same parameters. Under Time-0 Call price is about \$5.3451.

Next, we consider the same idea



$$C(S_{up}) \approx 7.9074, \qquad C(S_{down}) \approx 3.2064$$

So, repricing this, we have:

$$V_{up} = 1200(28.3475) - 1873(7.9074) + 1873(5.3451) \approx 29{,}217.77038.$$

$$V_{down} = 1200(20.9525) - 1873(3.2064) + 1873(5.3451) \approx 29{,}148.65497.$$

Therefore, the losses are such that:

$$Loss_{up} = 29{,}580 - 29{,}217.77038 = 362.23$$
$$Loss_{down} = 29{,}580 - 29{,}148.65497 = 431.35$$

So after delta-hedging, the portfolio losses are about \$360 if there is a rise of 15%, and a gain of about \$431 if intel falls 15%.

e)

In conclusion, the IBM slide's conclusion is the key idea that a portfolio can have zero sensitivity to the stock price but still have substantive risks. In this case, we are not fully delta-hedged, with some residual, but we do fundamentally minimise the risks.

Notably, "sensitivity' is only first-order, but options do introduce nonlinear P&L, such that the curvature is mainly gamma exposure, akin to the "convexity" term in fixed income.

For the scenario analysis, without delta-hedging with only 1,200 shares, we have a huge directional swing of:

- Up 15%: gain +\$4,437

- Down 15%: loss −$4,437

On the other hand, when delta-hedged, the risk is much smaller, when we have a loss of approximately either $360, or $430. As such zero-sensitivity does not mean low risk, it means that first-order sensitivity is neutralised at the present point, or in other owrds, the local, linear effects.

4)

4. **Monte Carlo VaR calculation**

A portfolio consisting of 100 shares of Apple and 120 share of IBM is purchased when the Apple price was $228.15 and the price of IBM was $205.23. Our empirical estimation of the Apple and IBM prices in 1 week gives the following 10 equally likely possibilities for the pair of prices:

ID	Apple	IBM
1	170.42	255.80
2	191.79	220.14
3	210.15	227.69
4	221.77	206.39
5	228.97	220.69
6	233.33	201.93
7	225.94	191.95
8	237.05	173.43
9	250.29	166.49
10	235.65	166.22

What are the 1 week 90th, 80th, 70th and 60th percentile VaR and ES of the portfolio?

	A	B	C	D	E	F	G
1			**ID**	**Apple**	**IBM**	**(V_1 = 100A+120I)**	**Loss (L=V_0-V_1)**
2	Px(Apple)	228.15	1	170.42	255.8	47,738.00	-295.40
3	Px(IBM)	205.23	2	191.79	220.14	45,595.80	1,846.80
4	Q(Apple)	100	3	210.15	227.69	48,337.80	-895.20
5	Q(IBM)	120	4	221.77	206.39	46,943.80	498.80
6	V_0	47442.6	5	228.97	220.69	49,379.80	-1,937.20
7			6	233.33	201.93	47,564.60	-122.00
8			7	225.94	191.95	45,628.00	1,814.60
9			8	237.05	173.43	44,516.60	2,926.00
10			9	250.29	166.49	45,007.80	2,434.80
11			10	235.65	166.22	43,511.40	3,931.20

Then a simple sorting will help, so now we have worst loss to profit represented as *Positives.*

B	C	D	E	F	G	H
	ID	Apple	IBM	(V_1 = 100A+12(	Loss (L=V_ V_1)	
228.15	10	235.65	166.22	43,511.40	3,931.20	
205.23	8	237.05	173.43	44,516.60	2,926.00	
100	9	250.29	166.49	45,007.80	2,434.80	
120	2	191.79	220.14	45,595.80	1,846.80	
47442.6	7	225.94	191.95	45,628.00	1,814.60	
	4	221.77	206.39	46,943.80	498.80	
	6	233.33	201.93	47,564.60	-122.00	
	1	170.42	255.8	47,738.00	-295.40	
	3	210.15	227.69	48,337.80	-895.20	
	5	228.97	220.69	49,379.80	-1,937.20	

As such, we consider the VaR, and note the point that 90% VaR as the 10% worst loss from slide 39:

Percentile	VaR
90^{th}	3,931
80^{th}	2,926
70^{th}	2,434.8
60^{th}	1,846.8

On the other hand, to calculate the ES, we have;

	A	B	C	D	E	F	G
1			ID	Apple	IBM	(V_1 = 100A+12(	Loss (L=V_ V_1)
2	Px(Apple)	228.15	10	235.65	166.22	43,511.40	3,931.20
3	Px(IBM)	205.23	8	237.05	173.43	44,516.60	2,926.00
4	Q(Apple)	100	9	250.29	166.49	45,007.80	2,434.80
5	Q(IBM)	120	2	191.79	220.14	45,595.80	1,846.80
6	V_0	47442.6	7	225.94	191.95	45,628.00	1,814.60
7			4	221.77	206.39	46,943.80	498.80
8			6	233.33	201.93	47,564.60	-122.00
9			1	170.42	255.8	47,738.00	-295.40
10			3	210.15	227.69	48,337.80	-895.20
11			5	228.97	220.69	49,379.80	-1,937.20
12							
13							
14							
15							
16							
17	Expected Shortfall						
18	90th ES	3,931.20	=AVERAGE(G2)				
19	80th ES	3,428.60	=AVERAGE(G2:G3)				
20	70th ES	3,097.33	=AVERAGE(G2:G4)				
21	60th ES	2,784.70	=AVERAGE(G2:G5)				

So from the simulations, we get the following numbers for ES:

Expected Shortfall (ES)	

90th ES	3,931.20
80th ES	3,428.60
70th ES	3,097.33
60th ES	2,784.70

For working out, please refer to attached excel file.

5)

5. **Formula VaR calculation**

Our portfolio is currently worth \$29000. We believe our portfolio follows GBM with drift $\mu = 0.040$ and volatility $\sigma = 35\%$. What is the 1 day, 1 week, and 1 year expected value of the portfolio value, standard deviation of the portfolio value and 98% VaR of the portfolio?

Note - 1 week is 5 trading days out of 252 trading days per year.

We note from the slides that:

EZ Mean of GBM

Alternatively, use martingales.

Define A_T by:

$$dA_T = \sigma A_T dW_T$$
$$A_T = e^{-\frac{\sigma^2}{2}T + \sigma W_T}$$
$$S_T = S_0 e^{\mu T} A_T$$

A_T is a martingale, so:

$$\begin{aligned} E[A_T] &= A_0 = 1 \\ E[S_T] &= E[S_0 e^{\mu T} A_T] \\ &= S_0 e^{\mu T} E[A_T] \\ &= S_0 e^{\mu T} \end{aligned}$$

This is why you should learn how to work with martingales!

EZ Variance of GBM

For the variance, we proceed analogously:

$$S_T = S_0 e^{(\mu - \frac{\sigma^2}{2})T + \sigma W_T}$$
$$S_T^2 = S_0^2 e^{(2\mu - \sigma^2)T + 2\sigma W_T}$$
$$dB_T = 2\sigma B_T dW_T$$
$$B_T = e^{-2\sigma^2 T + 2\sigma W_T}$$
$$S_T^2 = S_0^2 e^{(2\mu + \sigma^2)T} B_T$$

B_T is a martingale, so:

$$\begin{aligned} E[S_T^2] &= E[S_0^2 e^{(2\mu+\sigma^2)T} B_T] \\ &= S_0^2 e^{(2\mu+\sigma^2)T} E[B_T] \\ &= S_0^2 e^{(2\mu+\sigma^2)T} \\ \mathrm{var}[S_T] &= E[S_T^2] - E[S_T]^2 \\ &= S_0^2 e^{(2\mu+\sigma^2)T} - S_0^2 e^{2\mu T} \\ &= S_0^2 (e^{\sigma^2 T} - 1) e^{2\mu T} \end{aligned}$$

GBM VaR

Continuing...

$$P[S_T < X] = 1 - p$$
$$1 - p = \Phi\left(\frac{\log\left(\frac{X}{S_0}\right) - \left(\mu - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}\right)$$
$$\Phi^{-1}(1-p) = \frac{\log\left(\frac{X}{S_0}\right) - \left(\mu - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$
$$S_0 e^{\sigma\sqrt{T}\Phi^{-1}(1-p) + \left(\mu - \frac{\sigma^2}{2}\right)T} = X$$

This yields:

$$\mathrm{VaR}(S, T, p) = S_0 - S_0 e^{\sigma\sqrt{T}\Phi^{-1}(1-p) + \left(\mu - \frac{\sigma^2}{2}\right)T}$$

We, therefore, apply these formulas. It has been given that $V_0 = 29000$, $\mu = 0.040$, $\sigma = 0.35$, and then using the fact that T = 1 year, 1 week = 5/252, 1 day = 1/252.

For mean we have:

T = 1 day	T = 5/252	T = 1 year
$E[V_T] = V_0 e^{\mu T}$ $= 29{,}000 e^{(0.04)(1/252)}$ $= 29{,}004.60$	$E[V_T] = V_0 e^{\mu T}$ $= 29{,}000 e^{(0.04)(5/252)}$ $= 29{,}023.03$	$E[V_T] = V_0 e^{\mu T}$ $= 29{,}000 e^{(0.04)(1)}$ $= 30{,}183.51$

For the standard deviation:

T = 1 day	T = 5/252	T = 1 year
$SD(V_T) = V_0 e^{\mu T}\sqrt{e^{\sigma^2 T} - 1}$	$SD(V_T) = V_0 e^{\mu T}\sqrt{e^{\sigma^2 T} - 1}$	$SD(V_T) = V_0 e^{\mu T}\sqrt{e^{\sigma^2 T} - 1}$

$= 29{,}000e^{0.04(1/252)}\sqrt{e^{0.35^2(1/252)} - 1}$ $= 639.57$	$= 29{,}000e^{0.04(5/252)}\sqrt{e^{0.35^2(5/252)} - 1}$ $= 1{,}431.72$	$= 29{,}000e^{0.04(252/252)}\sqrt{e^{0.35^2(252/252)} -$ $= 10{,}896.17$

On the other hand, if we consider the VaR,

$$VaR_{0.98} = V_0\left[1 - exp\left(\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}z_{0.02}\right)\right],$$

$$z_{0.02} = \Phi^{-1}(0.02) \approx -2.0537$$

T = 1 day	T = 5/252	T = 1 year
VaR $= 29{,}000\Big[1$ $- exp\left(\left(0.04 - \frac{0.35^2}{2}\right)(1/252)\right.$ $\left.+ 0.35 * \sqrt{(1/252)}(-2.0537)\right)\Big]$ $VaR = 1{,}286.20$	VaR $= 29{,}000\Big[1$ $- exp\left(\left(0.04 - \frac{0.35^2}{2}\right)(5/252)\right.$ $\left.+ 0.35 * \sqrt{(5/252)}(-2.0537)\right)\Big]$ $VaR = 2{,}803.57$	VaR $= 29{,}000\Big[1$ $- exp\left(\left(0.04 - \frac{0.35^2}{2}\right)(1)\right.$ $\left.+ 0.35 * \sqrt{(1)}(-2.0537)\right)\Big]$ $VaR = 15{,}164.56$

6) See below.

Expected shortfall

The formula for expected shortfall for $V_T = S_T$ is more complicated to derive

$$\begin{aligned}\mathrm{ES}(S,T,p) &= -E^P[S_T - S_0 | S_T - S_0 < -\mathrm{VaR}(S,T,p)] \\ &= S_0 - E^P[S_T | S_T < S_0 - \mathrm{VaR}(S,T,p)]\end{aligned}$$

$$E^P[S_T | S_T < X] = \frac{\int_{-\infty}^{X} S_T \, dP}{\int_{-\infty}^{X} dP}$$

$$\int_{-\infty}^{X} S_T \, dP = \frac{1}{\sqrt{2\pi T}} \int_{-\infty}^{Y} S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma W} e^{-\frac{W^2}{2T}} dW$$

where Y satisfies $S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma Y} = X$.

Now, complete the square, solve for Y, and plug in $X = S_0 - \mathrm{VaR}(S,T,p)$, and note that with such an X, the denominator above is just $1 - p$.

Complete for homework!

6- Expected Shortfall Formula Derivation

By definition, $dS_t = \mu S_t dt + \sigma S_t dt$

We know that: $\quad W_T = Z \cdot \sqrt{T}, \; Z \sim N(0,1)$

$$S_T = \exp\left(\left(r - \frac{\sigma^2}{2}\right)T + \sigma W_T\right)$$

$L = S_0 - S_T$, such that:

$$ES(S,T,P) = \mathbb{E}[L \mid L \geq VaR(S,T,p)] = S_0 - \mathbb{E}[S_T \mid S_T < x],$$

where $x = S_0 - VaR(S,T,p)$, satisfies:

$$P(S_T < x) = 1 - p$$

Here, we find the cutoff, of "X",

We know $\log S_T$ is normal, so, we have

$$\log S_T = \underbrace{\log S_0 + (\mu + \tfrac{\sigma^2}{2})T}_{m} + \underbrace{\sigma\sqrt{T}}_{s} \cdot Z \quad , v = \sigma^2 T$$

Then:

$$\mathbb{P}(S_T < X) = \mathbb{P}(\ln S_T < \ln X) = P\left(Z < \frac{\ln X - m}{s}\right)$$

Then, we define z_{1-p} as the $(1-p)$-quantile, $N(0,1)$:

$z_{1-p} = \Phi^{-1}(1-p)$ ← CDF

By the VaR definition, we choose X, s.t $\mathbb{P}(S_T < X) = 1-p$, hence

$$\frac{\ln X - m}{s} = z_{1-p} \Rightarrow X = S_0 \exp\left((\mu - \tfrac{\sigma^2}{2})T + \sigma\sqrt{T} z_{1-p}\right)$$

cutoff, effectively.

So, also, we set $VaR(S,T,p) = S_0 - X$.

Then, we have:

$$\mathbb{E}[S_T \mid S_T < X] = \frac{\mathbb{E}[S_T \cdot \mathbb{1}_{\{S_T < X\}}]}{P(S_T < X)}$$

considering numerator + denominator separately.

Numerator:

$$\mathbb{E}[S_T \cdot \mathbb{1}_{\{S_T<K\}}] = \underbrace{S_0 e^{(\mu-\frac{\sigma^2}{2})T}} \cdot \underbrace{\mathbb{E}[e^{sZ} \cdot \mathbb{1}_{\{Z<z_{1-p}\}}]}$$

Then, consider the expectation

$$\cdot\ \mathbb{E}[e^{sZ} \cdot \mathbb{1}_{\{Z<z_{1-p}\}}] = \int_{-\infty}^{z} \underbrace{e^{sy}\phi(y)}\, dy, \quad \phi = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

From this, we complete the square

$$sy - \frac{1}{2}y^2 = \frac{1}{2}s^2 - \frac{1}{2}(y-s)^2,$$

So, from this, I set,

$$\int_{-\infty}^{z} e^{sy}\phi(y)dy = e^{s^2/2}\int_{-\infty}^{z} \cdot \frac{1}{\sqrt{2\pi}} e^{-(y-s)^2/2} dy$$

$$= e^{s^2/2}\, \Phi(z-s)$$

Therefore

$$\mathbb{E}[S_T \mathbb{1}_{\{S_T<K\}}] = S_0 e^{(\mu-\frac{1}{2}\sigma^2)T} \cdot e^{s^2/2}\, \Phi(z_{1-p}-s)$$

$$= S_0 e^{\mu T} \Phi(z_{1-p} - \sigma\sqrt{T})$$

Denominator

$$P(S_T < X) = \mathbb{P}(Z < z_{1-p}) = \Phi(z_{1-p}) = 1-p$$

$$\mathbb{E}[S_T \mid S_T < X] = S_0 e^{\mu T} \frac{\Phi(z_{1-p} - \sigma\sqrt{T})}{1-p}$$

Final Estimated Shortfall

$$ES(S,T,p) = S_0 - \mathbb{E}[S_T \mid S_T < X]$$

$$\boxed{ES(S,T,p) = S_0 \left[1 - e^{\mu T} \frac{\Phi(z_{1-p} - \sigma\sqrt{T})}{1-p} \right], \quad z_{1-p} = \Phi^{-1}(1-p)}$$

$$\Downarrow$$

$$\boxed{ES(S,T,p) = S_0 \left[1 - e^{\mu T} \frac{\Phi(\Phi^{-1}(1-p) - \sigma\sqrt{T})}{1-p} \right]}$$